U.S. Inflation Persistence Through Monthly and Annual Observations

QIDIAN XING

Executive summary

This study estimates a stationary latent AR(1) model of U.S. monthly inflation, then fits the same monthly process using overlapping twelve-month sums. Simulations with known parameters check the implementation before either empirical fit. An exact 24-state representation retains the observation shocks shared by adjacent annual windows. On 913 identical future-year targets, the annual fit has RMSE 2.20921 percentage points, compared with 2.38659 for the monthly fit and 2.43691 for a persistence benchmark. These comparisons use full-sample parameters and are retrospective.

The annual fit nevertheless implies lag-12 annual correlation 0.33895, below the historical 0.60718. Successful numerical checks therefore do not establish economic adequacy. A 200-replication Monte Carlo calibration at the historical sample length shows that model-based asymptotic standard errors are informative but not perfectly calibrated, particularly for the mean and persistence parameters.

The research question is how much monthly inflation reflects a persistent latent component, and whether observing the same process through overlapping annual sums improves its statistical description. Separating parameter recovery from historical model adequacy is essential to this comparison.

1. Data and research scope

The supplied CSV contains 936 consecutive months from January 1947 through December 2024. We use $y = 100 \times \text{Inflation}$, converting the raw decimal rate to monthly percentage points (pp). We do not multiply the monthly rate by 12, compound rates or alter the historical sample. The audit retains 138 zero and 116 negative observations; no observation is missing or duplicated.

The annual series is the sum of the latest twelve monthly rates. It contains 925 observations from December 1947 through December 2024 and receives a new observation each month. Its units are summed monthly percentage points, while Q and H below have squared-pp units.

Sample and forecast alignment

Series	Count	First date	Last date
Monthly observations	936	1947-01	2024-12
Rolling annual sums	925	1947-12	2024-12
Common forecast targets	913	1948-12	2024-12

2. A latent persistent component

Monthly inflation combines a constant mean, a persistent unobserved component and a transitory observation shock. We impose stationary persistence and independent Gaussian shocks:

$$y_t = \mu + \alpha_t + \eta_t, \qquad \alpha_{t+1} = \phi\alpha_t + \varepsilon_{t+1}$$

$$\eta_t \sim N(0, H), \quad \varepsilon_{t+1} \sim N(0, Q), \quad |\phi| < 1, \quad Q, H > 0$$

Both shock sequences are independent across time and of one another; the initial state is independent of observation shocks and future state shocks. The scalar observation and transition matrices are Z = R = 1, d = μ, T = φ and c = 0. The stationary variance solves V = φ²V + Q, which gives the prior used by simulation and every likelihood evaluation:

$$\alpha_1 \sim N\left(0, \frac{Q}{1-\phi^2}\right), \qquad a_{1|0} = 0, \quad P_{1|0} = \frac{Q}{1-\phi^2}$$

2.1 Score an observation before updating the state

The filter first forms a predictive mean and variance using past observations. After the current observation arrives, the innovation—the difference from that prediction—and its likelihood contribution are computed before updating the state:

$$v_t = y_t - \mu - a_{t|t-1}, \quad F_t = P_{t|t-1} + H, \quad K_t = \frac{P_{t|t-1}}{F_t}$$

$$a_{t|t} = a_{t|t-1} + K_t v_t, \quad P_{t|t} = (1 - K_t)P_{t|t-1}$$

$$a_{t+1|t} = \phi a_{t|t}, \quad P_{t+1|t} = \phi^2 P_{t|t} + Q$$

$$\ell = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log F_t + \frac{v_t^2}{F_t}\right]$$

Using the updated state to score the same observation would use that observation twice. The implementation follows the predictive-innovation-update order. Dense

joint-Gaussian likelihood comparisons provide an independent check of the recursion, including the stationary initialization.

3. Recover known parameters before using real data

The simulation truth is μ = 0.2, φ = 0.9, Q = 0.04 and H = 0.01. The initial state is a stationary draw, not a fixed zero. Separate, predeclared seeds generate samples of 300, 3,000 and 300,000 months; none is selected after inspecting its estimates. Figure 1 separates the observed level from its zero-mean state.

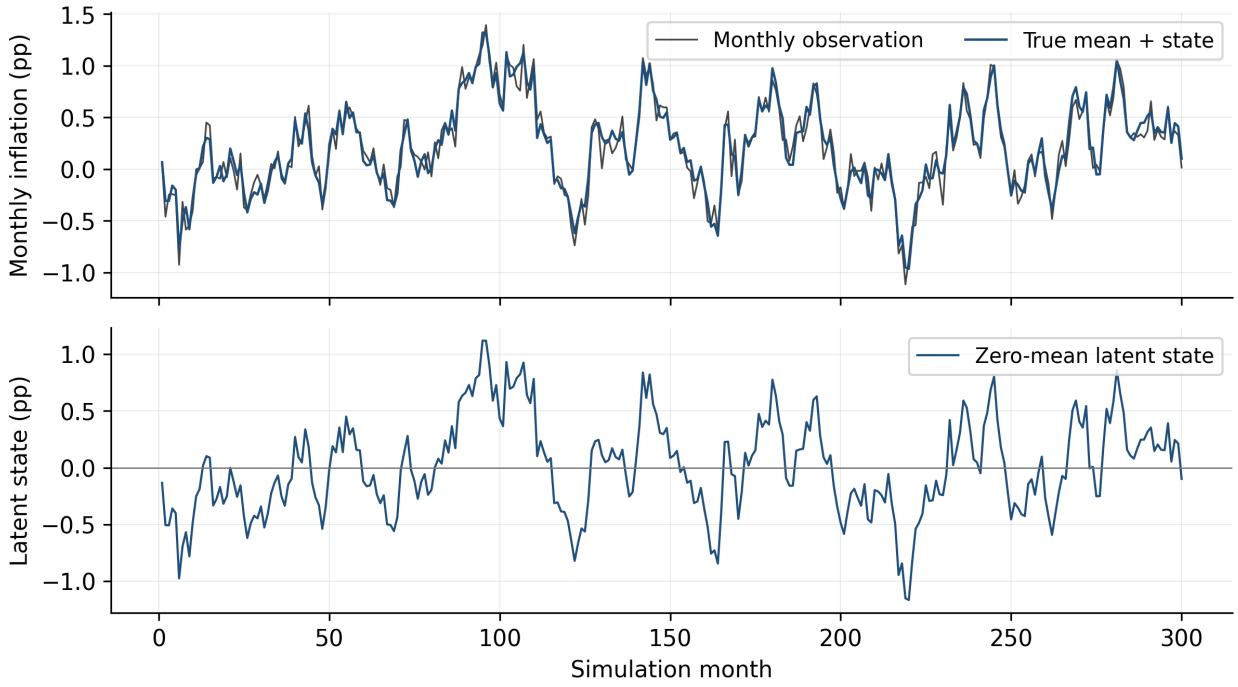


Figure 1. A 300-month draw (seed 431300). Observation noise separates the observed series from the true mean plus state; the lower panel removes the intercept.

Table 1. Monthly simulation estimates; model-based asymptotic SEs in parentheses. μ and shock SDs are in pp; φ is unitless.

Months	μ	φ	√Q	√H
True	0.20000	0.90000	0.20000	0.10000
300	0.19477 (0.08310)	0.85154 (0.03619)	0.21741 (0.01902)	0.06668 (0.03359)
3,000	0.18831 (0.03883)	0.90646 (0.00881)	0.19932 (0.00557)	0.10442 (0.00618)
300,000	0.19761 (0.00363)	0.89897 (0.00093)	0.20044 (0.00056)	0.10037 (0.00065)

Longer samples recover the true parameters more closely overall and sharply reduce uncertainty. An individual estimate need not improve monotonically. The 300-month sample gives a noisy estimate of observation variance; it remains in the results. Optimization uses four starts, and Appendix A explains the full-likelihood curvature and parameter transformation.

4. The historical monthly fit

The monthly estimate is $\mu = 0.28716$ pp, $\varphi = 0.75269$, with state and observation shock SDs 0.20431 and 0.23385 pp. Its one-year latent persistence, $\varphi^{12} = 0.03306$, is low. This measures the decay of a deviation in the monthly state; it does not directly describe the correlation of annual observations.

Table 2. Empirical estimates using monthly observations and rolling annual sums. Model-based asymptotic SEs appear in parentheses.

Observation fit	μ (pp)	φ	√Q (pp)	√H (pp)
Monthly	0.28716 (0.02799)	0.75269 (0.05379)	0.20431 (0.02725)	0.23385 (0.01972)
Annual sums	0.28740 (0.03421)	0.87010 (0.04624)	0.13252 (0.02753)	0.25965 (0.01538)

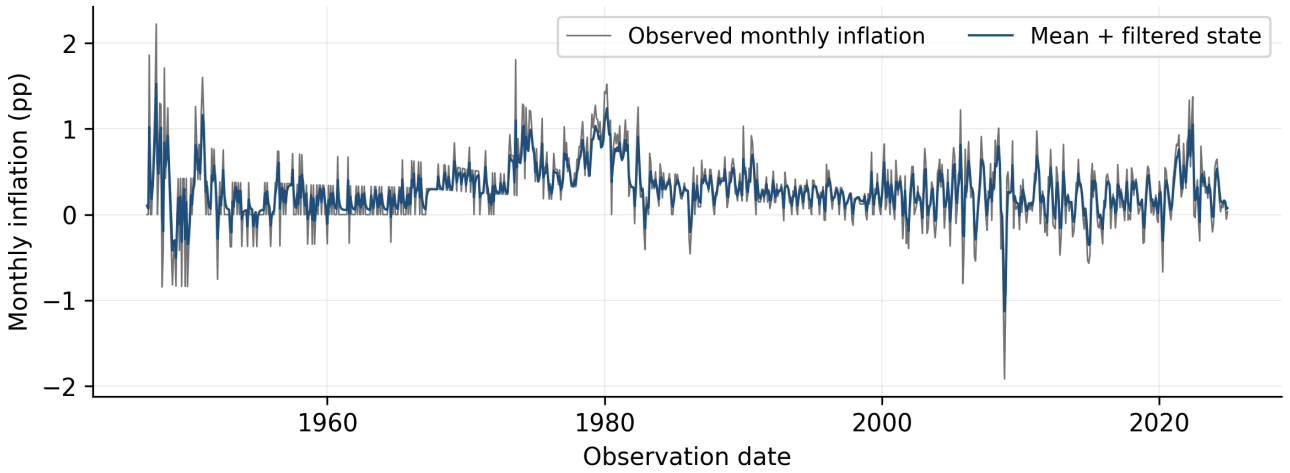


Figure 2. Monthly inflation and the estimated mean plus filtered state. The model attributes short-lived movements partly to observation noise.

4.1 Filtering describes what was known at a date

The filtered state conditions on observations through that date. The smoother also uses later observations and can reclassify an earlier movement as persistent or transitory. Both paths condition on full-sample parameter estimates. Smoothing is useful for historical description but cannot supply a forecast-origin state without future information.

4.2 Later observations revise the historical state

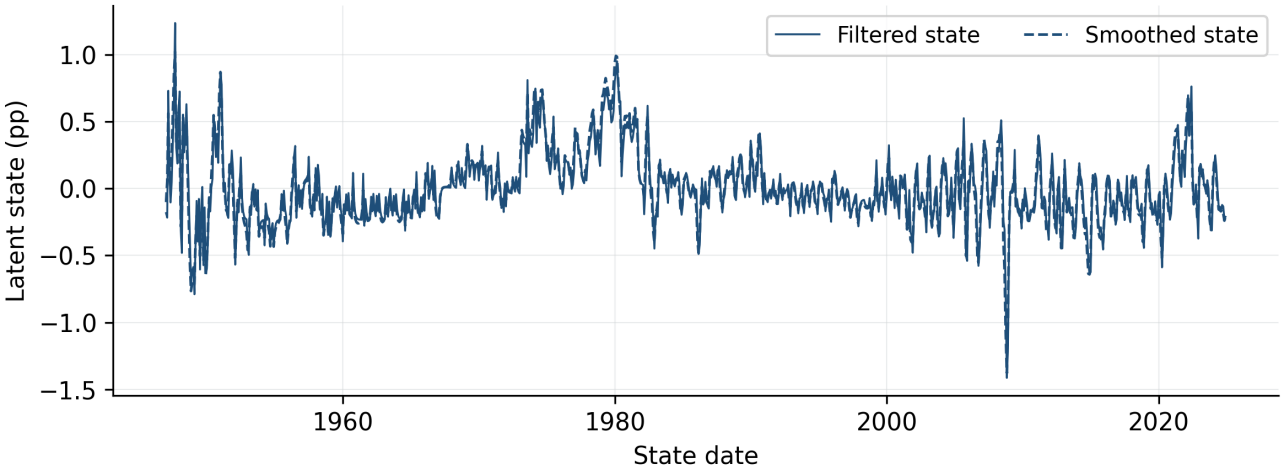


Figure 3. Filtered and smoothed zero-mean states. Differences reflect the smoother's access to later data; the two agree at the final date.

The Rauch–Tung–Striebel smoother works backward from the final filtered state. Its gain propagates subsequent revisions to earlier dates:

$$J_t = \frac{P_{t|t}\phi}{P_{t+1|t}}, \qquad a_{t|T} = a_{t|t} + J_t(a_{t+1|T} - a_{t+1|t})$$

$$P_{t|T} = P_{t|t} + J_t^2(P_{t+1|T} - P_{t+1|t})$$

At the terminal date, filtered and smoothed means and variances agree. Smoothing does not increase conditional variance. Both properties are verified, together with an independent dense Gaussian conditional-state calculation. These are implementation checks, not evidence that the model captures every economic regime.

A fixed-parameter h-step forecast uses the filtered state available at its origin and performs no updates with future observations:

$$a_{t+h|t} = \phi^h a_{t|t}, \qquad P_{t+h|t} = \phi^{2h} P_{t|t} + Q \sum_{j=0}^{h-1} \phi^{2j}$$

5. Rolling annual sums require a larger state

Aggregation can reduce the effect of a stable additive seasonal pattern that sums to zero over twelve months. Changing seasonal patterns need not cancel. More importantly, successive annual sums share eleven months. Their monthly observation shocks cannot be replaced by independent annual noise. We keep both lag blocks in the state:

$$z_t = \sum_{j=0}^{11} y_{t-j}, \qquad x_t = (\alpha_t, \ldots, \alpha_{t-11}, \eta_t, \ldots, \eta_{t-11})'$$

$$z_t = d^{aug} + Z^{aug} x_t, \qquad x_{t+1} = T^{aug} x_t + c^{aug} + R^{aug} e_{t+1}$$

$$Z^{aug} = \mathbf{1}_{1\times 24}, \quad d^{aug} = 12\mu, \quad H^{aug} = 0, \quad c^{aug} = \mathbf{0}_{24}$$

$$T^{aug} = \text{blockdiag}(A_\phi, A_0), \qquad Q^{aug} = \text{diag}(Q, H)$$

The two 12 × 12 transition blocks shift their lagged components down one row. For one-based indices i,j = 1,...,12, their complete entry definitions are:

$$(A_\phi)_{ij} = \phi\, \mathbf{1}_{i=j=1} + \mathbf{1}_{i=j+1}, \qquad (A_0)_{ij} = \mathbf{1}_{i=j+1}$$

The 24 × 2 loading matrix places the new state shock in the first row of the latent block and the new monthly observation shock in the first row of the noise block. Its only nonzero entries are:

$$R^{aug}_{1,1} = R^{aug}_{13,2} = 1, \qquad R^{aug}_{ij} = 0 \quad \text{otherwise}$$

The two-dimensional Gaussian shock has covariance Q^{aug}. The implementation constructs the transition, observation, loading and covariance matrices directly from the parameter values and verifies their dimensions and lag-shift structure.

5.1 A joint prior before the first annual observation

$$x_{12} \sim N(\mathbf{0}_{24}, P_0), \quad P_0 = \text{blockdiag}(\Sigma_\alpha, HI_{12})$$

$$(\Sigma_\alpha)_{ij} = \frac{Q}{1-\phi^2}\phi^{|i-j|}, \quad i,j = 1,\ldots,12$$

The two blocks have zero cross-covariance. This is the unconditional joint prior before observing z_{12}. Annual-only estimation does not initialize from monthly observations. $H^{aug} = 0$ means no additional annual measurement noise is added; it does not set the original monthly H to zero.

5.2 Validate the annual likelihood using simulated data

To validate the augmented likelihood before using historical data, we simulate 3,000 monthly observations from the known model and aggregate them into 2,989 rolling twelve-month sums. We fit their exact augmented likelihood before estimating the 925 historical sums. This checks that aggregation and the overlapping observation shocks are handled correctly while preserving the underlying monthly data-generating process.

Table 3. Annual-model validation using 2,989 sums of the 3,000-month simulation. SEs in parentheses.

Annual simulation	μ (pp)	φ	√Q (pp)	√H (pp)
Estimate	0.18826 (0.03880)	0.90628 (0.00883)	0.19956 (0.00559)	0.10444 (0.00621)

For the historical annual sums, φ rises to 0.87010, while the state-shock SD falls to 0.13252 pp. The observation-shock SD is 0.25965 pp. Annual latent persistence $\varphi^{12} = 0.18830$ is larger than the monthly fit's value. Different fits summarize the sample under different observation information sets; aggregation does not create a new structural inflation process.

Accumulated monthly observation noise retains the covariance created by overlapping annual windows:

$$u_t = \sum_{j=0}^{11} \eta_{t-j}, \quad \operatorname{Cov}(u_t, u_{t-k}) = (12-k)H, \quad k = 0, \ldots, 11$$

The covariance is zero at greater lags. The second state block retains that overlap. Dense Gaussian comparisons independently check the annual likelihood. Numerical covariance checks confirm positive innovation variances and no material negative filtered-covariance eigenvalues without adding jitter.

6. Forecast an entirely future year

Predicting the next monthly update of an annual sum shares eleven known months with the current sum. Instead, we predict the annual sum twelve months after the origin, whose twelve component months are all future observations. No intermediate update is allowed. The three predictions are:

$$\hat{z}^{m}_{t+12|t} = 12\hat{\mu}_m + a^{m}_{t|t} \sum_{h=1}^{12} \hat{\phi}^{h}_{m}$$

$$\hat{z}^{a}_{t+12|t} = 12\hat{\mu}_a + Z^{aug}(T^{aug})^{12} a^{aug}_{t|t}, \quad \hat{z}^{b}_{t+12|t} = z_t$$

All methods use origins December 1947–December 2023 and the same 913 targets December 1948–December 2024. The monthly model uses monthly observations through each origin; the annual model uses annual sums through that origin. The benchmark carries the current annual sum forward. Smoothed states are excluded.

6.1 The ranking depends on the loss function

Table 4. Errors on the same 913 future-year targets (pp).

Method	RMSE	MAE
Monthly	2.38659	1.76091
Annual	2.20921	1.60944
Benchmark	2.43691	1.67285

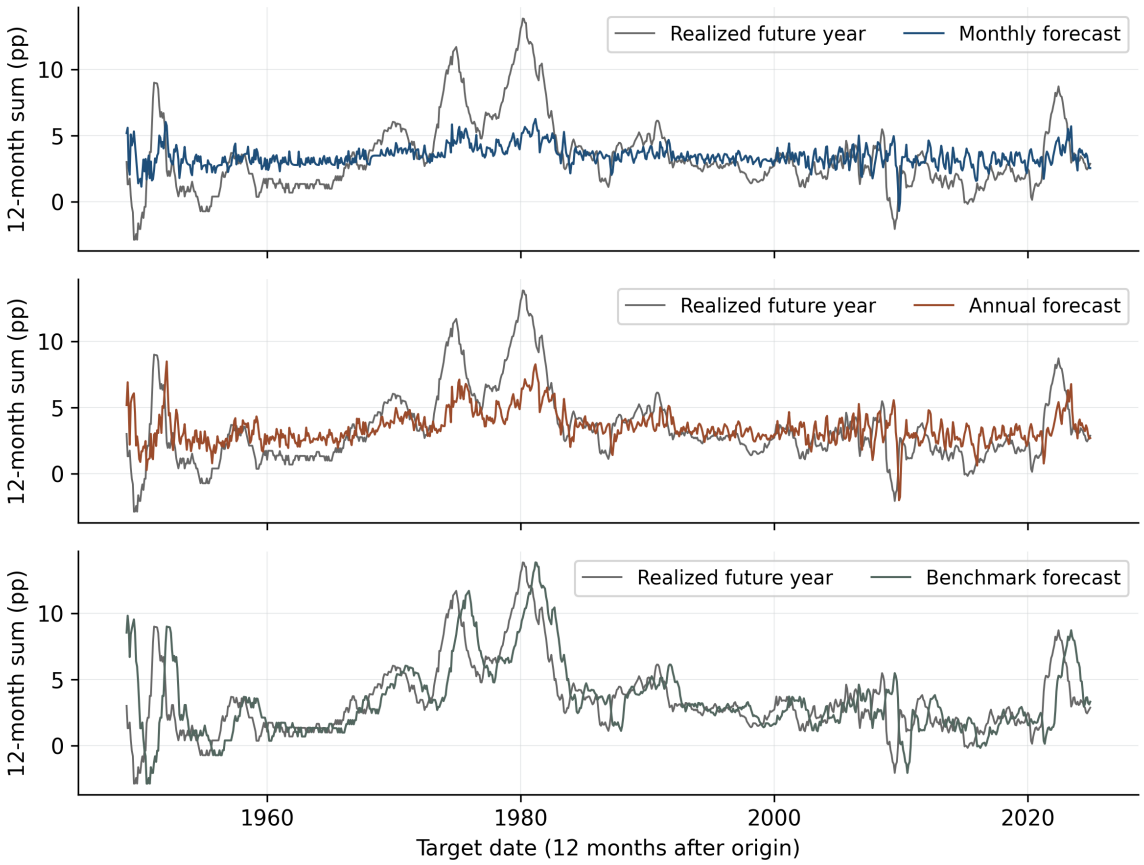


Figure 4. Forecasts aligned by target date, with common axes. Each panel compares its prediction with the same realized, entirely future twelve-month sum.

Relative to the monthly fit, the annual-observation fit reduces retrospective RMSE by 7.43%. This is a point comparison under full-sample parameter estimates.

The monthly fit has lower RMSE but higher MAE than the benchmark. Squared loss puts more weight on large errors, so this pattern is compatible with reducing some large misses while increasing typical absolute error. The annual fit has the lowest value of both metrics in this sample. We do not infer statistically significant superiority from these point comparisons.

Full-sample parameter estimates use information after early forecast origins. Although states respect their own origin information sets, this is a retrospective comparison, not a true out-of-sample forecast exercise. Overlapping evaluation windows also make forecast errors dependent. A formal predictive-accuracy test or real-time parameter exercise is outside the scope of this study.

7. Economic persistence remains underexplained

To check model implications beyond parameter recovery, we simulate four samples of 936 months from the annual fit's monthly parameters and convert each into 925 annual sums. Their seeds were predeclared. Figure 5 uses common axes so that differences in range and sustained movements remain visible.

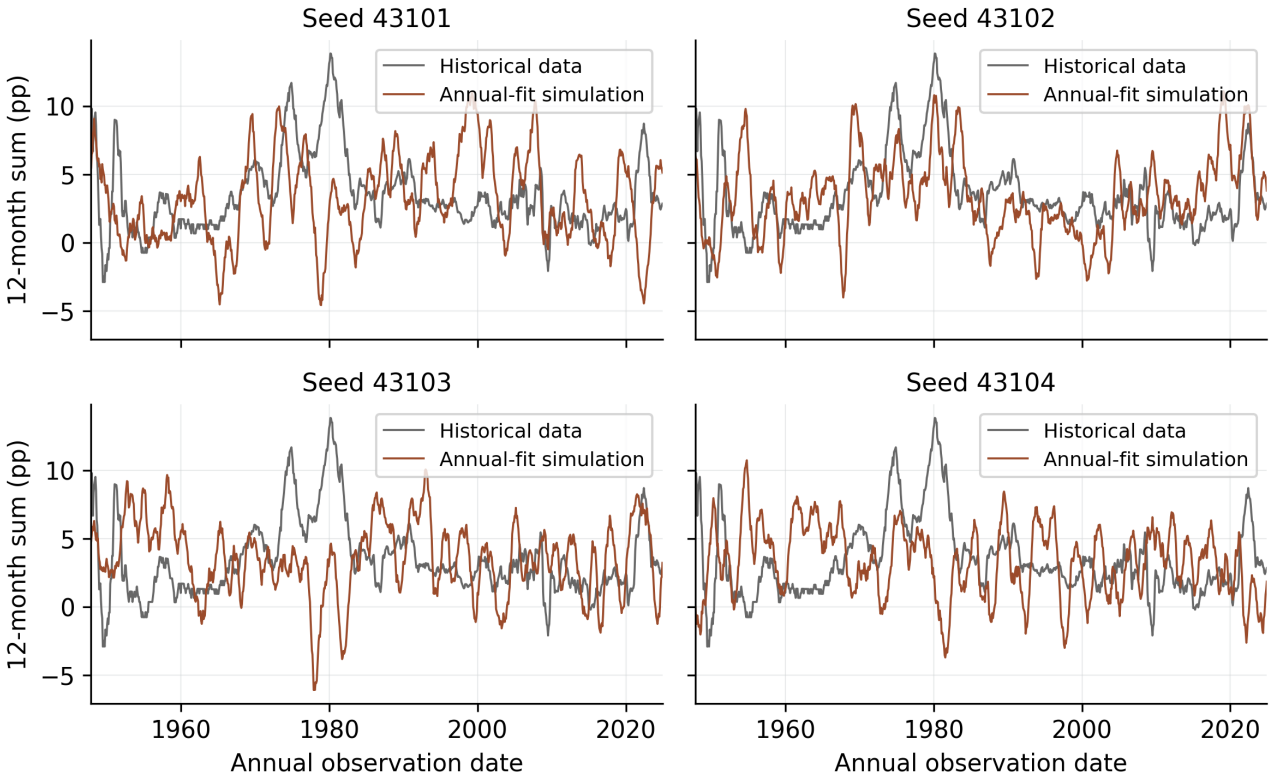


Figure 5. Historical annual sums and four independent annual-fit paths. Common scales reveal differences in sustained inflation episodes; simulations are not expected to reproduce historical events in the same years.

Table 5. Nonoverlapping annual persistence and variance.

Source	corr(z_t,z_{t-12})	φ^{12}	Var(z) (pp²)
Data	0.60718	—	7.61306
Monthly fit implied	0.16905	0.03306	6.55439
Annual fit implied	0.33895	0.18830	7.24653

The annual fit's implied correlation 0.33895 is closer to the data's 0.60718 than the monthly fit's 0.16905, but still falls well short. The model can generate considerable variation while reverting too quickly to a fixed mean. A stationary, homoskedastic Gaussian specification may miss persistent changes in levels or volatility; this descriptive comparison does not identify a causal mechanism.

7.1 Annual correlation differs from latent decay

The annual observations z_t and z_{t-12} share no months. Their covariance still sums the dependence across all pairs of component months. The analytic calculation is:

$$\gamma_y(k) = \frac{Q}{1-\phi^2}\phi^{|k|} + H\mathbf{1}_{k=0}$$

$$\gamma_z(k) = \sum_{i=0}^{11}\sum_{j=0}^{11}\gamma_y(k+j-i), \qquad \rho_z(12) = \frac{\gamma_z(12)}{\gamma_z(0)}$$

Thus φ^{12} describes latent monthly state decay, while ρ_z(12) describes observed annual sums and includes aggregation and the observation-noise contribution to variance. Treating them as interchangeable would obscure the economic diagnosis. Implementation tests establish that the specified model is computed correctly; comparison with historical persistence asks whether that specification is useful.

8. Monte Carlo calibration at the historical sample length

As an additional calibration check, we examine whether the reported Hessian standard errors describe repeated-sampling uncertainty at T = 936, the length of the historical monthly sample. We run B = 200 replications using the same true parameters as in the simulation exercise. Independent random streams are fixed in advance from master seed 20261003.

All 200 replications are retained: 200 produce valid parameter estimates and 200 produce valid standard errors, with 0 exceptions and 0 selected optimizer-status failures. Bias, empirical standard deviation, average reported standard error and coverage are calculated on the common set of replications for which all parameter standard errors are valid.

An optimizer status flag alone does not determine whether a replication enters the calibration summaries. Validity is evaluated using the gradient, interior parameter estimates, Hessian curvature and stability of the standard-error calculation. Companion summaries include every parameter-valid estimate, including any estimate without usable standard errors. All replication estimates, error messages and failure indicators are retained.

8.1 What the 200 replications say about SEs

Table 6. Sampling estimates and error calibration; n = 200 matched fits. All entries use full stored precision before rounding.

Parameter	True	Mean	Bias	SD	Mean SE
μ	0.20000	0.18361	-0.01639	0.06567	0.06248
φ	0.90000	0.89280	-0.00720	0.01874	0.01712
√Q	0.20000	0.20130	0.00130	0.00951	0.01014
√H	0.10000	0.09704	-0.00296	0.01224	0.01231

Table 7. SE/SD ratios and nominal 95% interval coverage.

Parameter	Mean SE / SD	Coverage	Coverage MCSE
μ	0.95136	0.9150	0.01972
φ	0.91368	0.9400	0.01679
√Q	1.06619	0.9400	0.01679
√H	1.00576	0.9700	0.01206

Average SEs are about 4.9% below empirical SD for μ and 8.6% below it for φ; shock-SD ratios are closer to one. Coverage ranges from 0.915 to 0.970. These finite-batch results qualify the asymptotic uncertainty statements rather than invalidate the likelihood implementation.

The bias estimate itself has Monte Carlo uncertainty SD/√n. Coverage has Monte Carlo uncertainty $\sqrt{[p(1-p)/n]}$, and the reported uncertainty of SD is a normal-reference approximation. Neither a ratio exactly equal to one nor coverage exactly equal to 0.95 is required. Conditioning on valid fits would matter if failures occurred; this run has none.

The batch mean for μ lies below truth. Its average raw sample mean is similarly low, so a single fixed batch cannot establish a systematic unconditional mean bias. A

sign-reflection test on the first predeclared sample checks Gaussian location symmetry without changing the random stream.

Conclusion

Annual observations improve the retrospective point-error metrics and imply greater persistence, but the fitted stationary process still understates nonoverlapping annual correlation. The main distinction is between a numerically verified likelihood and a model that adequately describes persistent historical inflation. The Monte Carlo calibration further shows why model-based asymptotic standard errors should be interpreted with finite-sample sampling uncertainty in mind.

Appendix A. Curvature, scaling and reproducibility

A.1 Full-likelihood Hessian and the delta method

We minimize average negative likelihood over transformed unconstrained parameters. Inference differentiates the full negative likelihood, including its parameter-dependent stationary prior. Equivalently, an average-likelihood Hessian would need multiplication by the observation count. We use neither the optimizer's approximate Hessian nor a pseudoinverse.

$$u = (\mu, b, \log Q, \log H), \quad \phi = \tanh b, \quad I_u = \nabla_u^2[-\ell(u)]$$

$$g = (\mu, \phi, \sqrt{Q}, \sqrt{H}), \quad J = \operatorname{diag}(1, 1-\phi^2, \sqrt{Q}/2, \sqrt{H}/2)$$

$$\operatorname{Cov}(\hat{g}) = JI_u^{-1}J', \quad \operatorname{SE}(\hat{g}_j) = \sqrt{[JI_u^{-1}J']_{jj}}$$

Variance-parameter SEs use the analogous Jacobian diag(1,1−φ²,Q,H). Three central-difference step scales—0.002, 0.001 and 0.0005, adjusted by coordinate magnitude—check curvature. The middle scale supplies the covariance; every fit must have positive Hessian eigenvalues, average gradient below 10^{-6} and relative SE variation below 2%. These are model-based asymptotic Hessian/delta-method SEs, not misspecification-robust estimates.

Table A1. Full log likelihood and selected numerical diagnostics. Fit diagnostics retain full start records and Hessian matrices.

Fit	Log likelihood	Avg. gradient	Min. eigenvalue	SE variation
Monthly T=300	9.68560	3.25e-09	9.60e-01	2.00e-05
Monthly T=3,000	15.83471	1.20e-09	6.07e+01	3.81e-06
Monthly T=300,000	4159.96185	4.74e-10	5.23e+03	3.88e-06
Monthly data	-305.10472	5.80e-10	9.30e+00	1.16e-05
Annual simulation	-20.15607	1.55e-09	6.01e+01	3.80e-06
Annual data	-285.05560	2.54e-09	4.66e+00	2.17e-05

The eigenvalue and step checks support local numerical reliability. They do not prove that every likelihood has a unique global maximum, that Gaussian assumptions are correct or that an asymptotic interval is exact at finite T.

A.2 Reproduce the research results

Clone the public repository MartinXQD/us-inflation-state-space-model and follow README.md to create a project-local Python 3.12 environment and install the pinned dependencies. The verified Apple Silicon installation uses the documented OpenBLAS wheel branch. Other platforms have not been clean-install verified. From the repository root, run python scripts/run_all_notebooks.py. The runner executes six independent kernels and saves their outputs. For interactive work, start a kernel in notebooks/ and use Restart Kernel / Run All. Notebooks 05 and 06 estimate their own required fits; Notebook 02 recomputes all three simulation lengths and every one of the 200 Monte Carlo attempts. Only the unchanged CSV in data/ is read; no saved external results or earlier notebook variables are needed.

Data and methodological provenance

The empirical analysis uses the supplied US_inflation47_25.csv snapshot, covering January 1947 through December 2024 despite its filename. The data and initial model specification originated in an MFE course setting. Upstream dataset provenance and a separate redistribution license were not specified in the supplied materials; no replacement or external dataset was used. Model notation follows the supplied teaching notes on linear Gaussian state-space models and Kalman filtering. All numerical tables and figures are computed by this implementation. AI assistance and material explanation corrections are summarized in docs/AI_Assistance_and_Validation.md in the public repository.